SIMATS ENGINEERING

ASSESSMENT TOOL 2: Scenario-Based Research Assessment

COURSE NAME: ARTIFICIAL INTELLIGENCE AND EXPERT SYSTEMS

COURSE OUTCOME COVERED
COURSE CODE: MLA01

CO1: *Apply search algorithms and heuristic techniques to solve state-space search and optimization problems.* (BTL 3)

Assessment Tool Weightage: 45 %

Total Marks: 50 Time: 60 Mins No. of Questions: 5

Annexure A

Scenario-Based Research Assessment

Code of Conduct:

I ______________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Criterion	Weightage	Marks Obtained
Identification of Latest AI Application	10%	
Problem Analysis and Domain Understanding	15%	
AI Technologies and Working Mechanism	20%	
Research Quality and Literature Review	10%	
Real-World Implementation and Case Analysis	10%	
Impact, Challenges, and Future Scope	15%	
Originality and Critical Thinking	10%	
Organization, Presentation, and Documentation	10%	

Total 100%

Signature of the Faculty Total Marks Awarded

ANSWER ANY ONE

No.	AI Domain	Scenario-Based Research Question
1	**Healthcare**	Healthcare organizations are increasingly adopting Artificial Intelligence to improve diagnosis, treatment, and patient monitoring. Identify and research a latest AI-based application in healthcare (such as AI-assisted medical diagnosis, AI radiology systems, robotic surgery, AI drug discovery, or virtual healthcare assistants). Analyze its working principles, AI techniques used, real-world implementation, benefits, and limitations.
2	**Finance**	Financial institutions are leveraging Artificial Intelligence to enhance security, decision-making, and customer services. Identify and research a latest AI application in the finance domain (such as AI fraud detection systems, AI-powered financial advisors, algorithmic trading platforms, or intelligent banking assistants). Discuss its functionality, technologies involved, advantages, and challenges.
3	**Retail**	Retail companies are adopting AI-driven solutions to provide personalized and intelligent shopping experiences. Identify and research a latest AI application in retail (such as Generative AI shopping assistants, AI recommendation engines, virtual try-on systems, or AI-based demand forecasting). Explain its application, AI methods used, impact on customers and businesses, and future scope.
4	**Autonomous Vehicles**	The transportation sector is rapidly evolving with AI-based autonomous systems. Identify and research a latest AI application in autonomous vehicles (such as self-driving technology, AI-based perception systems, autonomous delivery vehicles, or intelligent traffic management). Analyze the AI technologies involved, real-world deployment, benefits, and challenges.
5	**Robotics**	Intelligent robots are being developed for applications in industries, healthcare, education, and daily life. Identify and research a latest AI application in robotics (such as humanoid robots, collaborative robots, service robots, or AI-powered warehouse robots). Explain their capabilities, AI techniques, real-world usage, and future developments.
6	**Manufacturing**	Manufacturing industries are integrating AI to achieve smart and automated production environments. Identify and research a latest AI application in smart manufacturing (such as AI-based predictive maintenance, computer vision quality inspection, digital twins, or autonomous manufacturing systems). Discuss its working, industrial applications, advantages, and limitations.

MLA0103-ARTIFICIAL INTELLIGENCE WITH EXPERT SYSTEMS

ASSIGNMENT TOOL

Name: Tejaswini(192525171)

Department: B.Tech AIML

Faculty name: Dr. M. Prakash

1. Title of AI Application

AI-Powered Autonomous Robot Vacuum Cleaner (iRobot Roomba j9+)

AI Domain: Robotics

Real-World Product: iRobot Roomba j9+

2. Domain Selection

Selected Domain: Robotics

Robotics is an AI application domain where intelligent machines perform tasks automatically with minimal human intervention. Modern robots use Artificial Intelligence to perceive their surroundings, make decisions, and perform actions safely and efficiently.

3. Introduction and Background

The **iRobot Roomba j9+** is one of the world's most advanced AI-powered robotic vacuum cleaners. Unlike traditional vacuum cleaners that require manual operation, the Roomba uses AI, computer vision, sensors, and machine learning to clean homes autonomously.

It creates a digital map of the house, identifies furniture and obstacles, avoids objects like shoes and pet waste, and cleans rooms efficiently without human guidance. The robot continuously improves its navigation through AI learning.

As smart homes become increasingly popular, AI-powered robots like Roomba improve convenience, save time, and reduce manual household work.

4. Problem Identification

Traditional vacuum cleaning has several challenges:

- Requires manual effort.
- Time-consuming for busy people.
- Difficult to clean under beds, sofas, and furniture.
- Inefficient cleaning due to repeated paths.
- Elderly and physically disabled people may find cleaning difficult.

These problems increase household workload and reduce cleaning efficiency.

5. Need for AI Solution

Artificial Intelligence is required because a robot must understand its surroundings before moving safely.

Compared to traditional robotic vacuum cleaners that move randomly, AI enables the robot to:

- Recognize obstacles.
- Plan the shortest cleaning path.
- Avoid collisions.
- Detect dirty areas requiring extra cleaning.
- Learn the house layout over time.
- Resume cleaning automatically after charging.

AI makes cleaning faster, smarter, and more efficient.

6. AI Technologies Used

The Roomba uses several AI technologies:

- **Machine Learning (ML):** Learns room layouts and user cleaning preferences.
- **Computer Vision:** Identifies furniture, cables, shoes, pet bowls, and obstacles.
- **Deep Learning:** Recognizes different household objects.
- **SLAM (Simultaneous Localization and Mapping):** Builds a map while tracking its location.

- **Sensor Fusion:** Combines data from cameras, infrared sensors, cliff sensors, and wheel encoders.
- **Path Planning Algorithms:** Finds the most efficient cleaning route.

7. Architecture Diagram

AI-Powered Robot Vacuum Cleaner

(irobot Roomba j9+)

User / Mobile App

(Cleaning Schedule, Commands)

Input Layer

• RGB Camera

• Infrared Sensors

• Cliff Sensors

• Dirt Detection Sensor

• Wheel Encoders

• Battery Sensor

Data Processing Layer

• Image Processing

• Sensor Data Collection

• Data Cleaning

• Sensor Fusion

• Feature Extraction

AI Intelligence Layer

• Machine Learning

• Deep Learning

• Computer Vision

• SLAM (Localization & Mapping)

• Object Recognition

Decision-Making Layer

• Path Planning

• Obstacle Avoidance

• Room Mapping

• Navigation Control

• Cleaning Optimization

Robot Control Layer

• Motor Controller

• Wheel Controller

• Vacuum Controller

• Battery Management

Output Layer

• Autonomous Cleaning

• Smart Navigation

• Auto Dock Charging

• Cleaning Status Report

• Mobile App Notification

8.Architecture Components

Layer	Components
Input Layer	Camera, Infrared Sensor, Cliff Sensor, Dirt Sensor, Wheel Encoder, Battery Sensor
Data Processing Layer	Data Cleaning, Image Processing, Sensor Fusion, Feature Extraction
AI Intelligence Layer	Machine Learning, Deep Learning, Computer Vision, SLAM, Object Recognition
Decision-Making Layer	Path Planning, Obstacle Avoidance, Navigation, Cleaning Optimization
Robot Control Layer	Motor Control, Wheel Control, Vacuum Control, Battery Management
Output Layer	Floor Cleaning, Auto Charging, Cleaning Report, Mobile Notification

9. Working Mechanism

Step 1: Input

The robot receives information from:

- Camera
- Infrared sensors
- Cliff sensors
- Dirt detection sensors

- Wheel encoders
- Battery sensors

Step 2: Data Processing

The AI system:

- Captures images.
- Measures distances.
- Detects walls and furniture.
- Identifies obstacles.
- Creates a digital house map.

Step 3: AI Decision Making

The AI decides:

- Which room to clean first.
- Which path is shortest.
- Which areas are already cleaned.
- Whether an obstacle should be avoided.
- When to return to the charging dock.

Step 4: Output

The robot:

- Cleans the floor.

- Avoids obstacles.
- Returns to charging automatically.
- Sends cleaning reports to the mobile app.

10. Data Sources and Processing

Data Sources

- Camera images
- Infrared sensor readings
- Cliff sensor data
- Dirt detection sensor data
- Wheel movement data
- User cleaning schedules
- Mobile app commands

Data Processing

1. Collect sensor data.
2. Clean and combine the data.
3. Generate a map using SLAM.
4. Detect objects using Computer Vision.
5. Select the best cleaning path.
6. Execute cleaning.
7. Store map updates for future cleaning.

11. Real-World Implementation (Use Case)

Organization

iRobot Corporation

Use Case

A working professional leaves home at 9:00 AM.

Using the mobile app, they schedule the Roomba to clean every morning.

The robot:

- Maps the entire house.
- Cleans each room systematically.
- Avoids shoes, toys, and pet waste.
- Detects heavily soiled areas and cleans them again.
- Returns to the charging dock automatically.
- Sends a cleaning completion notification to the user's smartphone.

Other Places Where It Is Used

- Smart homes
- Apartments
- Offices
- Hotels
- Hospitals
- Small commercial buildings

12. Impact and Benefits

The AI-powered robot provides:

- Saves time.
- Reduces manual effort.
- Intelligent navigation.
- Better cleaning efficiency.
- Lower energy consumption.
- Automatic scheduling.
- Improved user convenience.
- Safe obstacle avoidance.
- Continuous learning for better performance.

13. Challenges and Limitations

Technical Challenges

- Difficulty recognizing transparent objects.
- Limited battery life.
- Performance depends on lighting conditions.
- Complex room layouts may reduce efficiency.

Ethical & Privacy Challenges

- Indoor camera privacy concerns.
- Storage of home mapping data.

- Cybersecurity risks for connected devices.

Deployment Challenges

- High purchase cost.
- Regular maintenance required.
- Needs stable Wi-Fi for smart features.
- May struggle with very thick carpets or cluttered spaces.

14. Future Scope and Emerging Trends

Future robotic vacuum cleaners may include:

- More accurate AI object recognition.
- Voice interaction using Large Language Models (LLMs).
- Multi-floor intelligent mapping.
- Better battery technology.
- Swarm robotics (multiple robots working together).
- Integration with smart home ecosystems.
- Predictive cleaning based on user habits.
- Improved edge AI for faster onboard decision-making.

15. Conclusion

The **iRobot Roomba j9+** demonstrates how Artificial Intelligence has transformed household robotics. By combining Machine Learning, Computer Vision, SLAM, and intelligent navigation, it performs cleaning tasks efficiently with minimal human intervention. This real-world robotic application improves convenience, reduces manual labor, and represents the growing role of AI in everyday life. As AI technology advances, service robots are expected to become even smarter, more autonomous, and more capable of assisting people in homes and workplaces.

16. References

1. **Arjun, M., Oliver, C. A., & Joseph, A. M. (2024).** *Robot Vacuum Cleaner Using SLAM and ROS.* **IEEE International Conference on Knowledge Engineering, Communication Systems and Technology (ICKECS 2024).** DOI: 10.1109/ICKECS61492.2024.10616915. This paper presents an autonomous robot vacuum cleaner using ROS and SLAM for efficient indoor navigation.
2. **iRobot Corporation.** *Roomba j9+ Robot Vacuum Product Information.* Official documentation describing AI-based obstacle avoidance, mapping, and smart cleaning features. iRobot Official Website
3. **Al-Tawil, B., Hempel, T., Abdelrahman, A., & Al-Hamadi, A. (2024).** *A Review of Visual SLAM for Robotics: Evolution, Properties, and Future Applications.* **Frontiers in Robotics and AI**, 11. DOI: 10.3389/frobt.2024.1347985. This paper reviews modern Visual SLAM techniques widely used in intelligent mobile robots such as robotic vacuum cleaners.

Rubrics for Evaluation

S.No	Criteria	Weightage (%)	Excellent (100–90%)	Good (89–75%)	Satisfactory (74–60%)	Improvement (
1.	Identification of Latest AI Application	10%	Selects a highly relevant, emerging AI application with strong justification.	Selects a recent and relevant AI application.	Selects a suitable application with limited justification.	Application is outdated or irrelevant.
2.	Problem Analysis and Domain Understanding	15%	Clearly explains the real-world problem, domain context, and need for AI with critical analysis.	Good explanation with adequate analysis.	Basic explanation with limited analysis.	Problem statement is unclear or incomplete.
3.	AI Technologies and Working Mechanism	20%	Accurately explains AI techniques, algorithms, architecture, and workflow with technical depth.	Good explanation with minor omissions.	Covers only basic AI concepts.	AI technologies are poorly explained or incorrect.
4.	Research Quality and Literature Review	10%	Uses recent, credible sources with proper citations and excellent synthesis.	Good quality references with appropriate citations.	Limited references with basic discussion.	Few or unreliable references; poor citation.
5.	Real-World Implementation and Case Analysis	10%	Thoroughly analyzes industrial implementation with appropriate examples.	Good implementation analysis.	Basic implementation discussion.	Minimal or no real-world analysis.
6.	Impact, Challenges, and Future Scope	15%	Critically evaluates benefits, limitations, ethical issues, and future trends.	Good evaluation with minor gaps.	Basic discussion of impact and future scope.	Limited evaluation or missing sections.
7.	Originality and Critical Thinking	10%	Demonstrates independent research, innovation, and critical evaluation.	Good analytical thinking.	Some originality with limited analysis.	Mostly descriptive or copied content.
8.	Organization, Presentation, and Documentation	10%	Exceptionally organized, professional presentation, error-free, and properly referenced.	Well organized with minor issues.	Adequately organized with some errors.	Poor organization and documentation.